

# Host Galaxies of Type 1 AGNs

Zhijun Zhang

August 15, 2021

# Data Reduction and Image Decomposition

- Data reduction including sky subtraction and PSFs.
- Decomposing and fitting the galaxy images using GALFIT.
- Sérsic profile:

$$\mu(r) = \mu_e e^{-\kappa[(r/R_e)^{1/n} - 1]}$$

- Employ Fourier modes to characterize asymmetry, which indicates interactions or mergers.

## Derived Properties

- AGN and host galaxy luminosities and spectrum.
- Virial BH masses:

$$M_{\text{BH}} = \frac{fv^2 R_0}{G}$$

where  $R_0$  is the broad line region (BLR) in AGN. The BLR is photoionized by the central continuum source, and it is gravitationally bound to the SMBH. The normalization coefficient  $f = 1.31$  for  $v$  set to the FWHM of  $H\beta$ .

- Buldge type classification using the Sérsic index and bulge-to-total light ratio.

# Statistical Trends

- Host galaxies of Radio-loud ( $R = f_{6\text{cm}}/f_{4400\text{\AA}} \geq 10$ ) AGNs are brighter and more bulge-dominated than their radio-quiet counterparts.
- More disk dominance, higher fraction of pseudo-bulges, higher bar frequency, and lower merger fraction implicate secular processes as the main agent for BH growth in NLS1s.
- ? Merging features

# Conclusion

- $H\beta$  narrow line Type 1 AGNs prefer to host in later-type, lower-luminosity galaxies.
- The fraction of AGN hosts shows that morphological signatures of tidal interactions merger signatures is larger for more luminous hosts.
- RL AGNs generally preferentially live in earlier-type, more massive hosts, but not specially interacting or merger systems.

# Discussion

- Does dark matter distribution affect the  $B/T$  measurement?
- What secular process does this article discuss?
- Why do the authors think that the trend about buldge dominating is precisely the opposite on Page 16?

Thank you for listening!